

Balkanpharma Levofloxacin 5 mg/ml eye drops solution

Product Description

Clear liquid with pale yellow to pale greenish-yellow colour

Active Ingredient

1ml of eye drops solution, contains 5.12mg of levofloxacin hemihydrate equivalent to 5mg of levofloxacin

Excipients

benzalkonium chloride (preservatives)
sodium chloride
hydrochloric acid and/or sodium hydroxide (for pH adjustment)
water for injections

Dosage Form Eye drops, solution

Pharmacodynamics

Pharmacotherapeutic group: Ophthalmologicals, antiinfectives, fluoroquinolones
ATC Code: S01AE05

Levofloxacin is the L-isomer of the racemic drug substance ofloxacin. The antibacterial activity of ofloxacin resides primarily in the L-isomer.

Mechanism of action

As a fluoroquinolone antibacterial agent, levofloxacin inhibits bacterial type II topoisomerases - DNA gyrase and topoisomerase IV. Levofloxacin preferentially targets DNA gyrase in Gram-negative bacteria and topoisomerase IV in Gram-positive bacteria.

Mechanisms of resistance

Bacterial resistance to levofloxacin can develop primarily due to two main mechanisms, namely a decrease in the intrabacterial concentration of a drug, or alterations in a drug's target enzymes. Target site alteration results from mutations in the chromosomal genes encoding the DNA gyrase (*gyrA* and *gyrB*) and topoisomerase IV (*parC* and *parE*; *grlA* and *grlB* in *Staphylococcus aureus*). Resistance due to low intrabacterial drug concentration follows either from altered outer-membrane porins (OmpF) leading to reduced entry of fluoroquinolones in Gram-negative bacteria or from efflux pumps. Efflux-mediated resistance has been described in pneumococci (PmrA), staphylococci (NorA), anaerobes, and Gram-negative bacteria. Finally, plasmid-mediated resistance to quinolones (determined by the *qnr* gene) has been reported in *Klebsiella pneumoniae* and in *E. coli*.

Cross-resistance

Cross-resistance between fluoroquinolones may occur. Single mutations may not result in clinical resistance, but multiple mutations generally do result in clinical resistance to all drugs within the fluoroquinolone class. Altered outer-membrane porins and efflux systems may have a broad substrate specificity, targeting several classes of antibacterial agents and leading to multiresistance.

Breakpoints

MIC breakpoints separating susceptible from intermediately susceptible organisms and intermediately susceptible from resistant organisms according to breakpoint of EUCAST (European Committee on Antimicrobial Susceptibility Testing) are as follows:

Pseudomonas spp., *Staphylococcus* spp., *Streptococcus* A,B,C,G:

Susceptible $\leq$ 1 mg/L, resistant $>$ 2 mg/L

Streptococcus pneumoniae: Susceptible $\leq$ 2 mg/L, resistant $>$ 2 mg/L

Haemophilus influenzae, *Moraxella catarrhalis*: Susceptible $\leq$ 1 mg/L, resistant $>$ 1 mg/L

All other pathogens: Susceptible $\leq$ 1 mg/L, resistant $>$ 2 mg/L

Antibacterial spectrum

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. Therefore the information presented provides only an approximate guidance on probabilities as to whether microorganisms will be susceptible to levofloxacin or not. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Only those bacterial species that are commonly responsible for external ocular infections, such as conjunctivitis, are presented here in the following table.

Antibacterial spectrum – susceptibility category and resistance characteristics according to EUCAST

Category I: Commonly susceptible species
Aerobic Gram-positive micro-organisms

<i>Staphylococcus aureus</i> (MSSA)*	
<i>Streptococcus pneumoniae</i>	
<i>Streptococcus pyogenes</i>	
Viridans group streptococci	
Aerobic Gram-negative micro-organisms	
<i>Escherichia coli</i>	
<i>Haemophilus influenzae</i>	
<i>Moraxella catarrhalis</i>	
<i>Pseudomonas aeruginosa</i>	(Community isolates)
Other micro-organisms	
<i>Chlamydia trachomatis</i>	(Treatment of patients with chlamydial conjunctivitis requires concomitant systemic antimicrobial treatment)
Category II: Species for which acquired resistance may be a problem	
Aerobic Gram-positive micro-organisms	
<i>Staphylococcus aureus</i> (MRSA)**	
<i>Staphylococcus epidermidis</i>	
Aerobic Gram-negative micro-organisms	
<i>Pseudomonas aeruginosa</i>	(Hospital isolates)

* MSSA = methicillin-susceptible strains of *Staphylococcus aureus*

** MRSA = methicillin-resistant strains of *Staphylococcus aureus*

Resistance data presented in the table are based on the results of a multicentre surveillance study (Ophthalmic Study) on the prevalence of resistance among bacterial isolates obtained from patients with eye infections in Germany, June –November 2004.

Organisms have been classified as levofloxacin-susceptible based on in-vitro susceptibility and plasma concentrations reached after systemic therapy. Topical therapy achieves higher peak concentrations than found in plasma. However, it is not known if or how the kinetics of the drug after topical application to the eye may modify the antibacterial activity of levofloxacin.

Paediatric population

Pharmacodynamics properties are the same in adults and children aged ≥ 1 year.

Pharmacokinetics

After ocular instillation, levofloxacin is well maintained in the tear-film.

In a healthy-volunteer study, mean tear-film concentrations of levofloxacin measured four and six hours after topical dosing were 17.0 and 6.6 $\mu\text{g/mL}$, respectively. Five of six subjects studied had concentrations of 2 $\mu\text{g/mL}$ or above at 4 hours post dose. Four of the six subjects maintained this concentration at 6 hours post dose.

Levofloxacin concentration in plasma was measured in 15 healthy adult volunteers at various time points during a 15-day course of treatment with Levofloxacin 5 mg/ml eye drops, solution. The mean levofloxacin concentration in plasma 1 hour post-dose ranged from 0.86 ng/mL on Day 1 to 2.05 ng/mL on Day 15. The highest maximum levofloxacin concentration of 2.25 ng/mL was measured on Day 4 following 2 days of dosing every 2 hours for a total of 8 doses per day. Maximum levofloxacin concentrations increased from 0.94 ng/mL on Day 1 to 2.15 ng/mL on Day 15, which is more than 1000 times lower than those reported after standard oral doses of levofloxacin.

As yet, the plasma concentrations of levofloxacin reached after application to infected eyes are not known.

Indication

For the treatment of bacterial conjunctivitis caused by susceptible strains of the designated microorganisms.

Recommended Dose

Usually, instill 1 drop a time 3 times daily. The dosage may be adjusted according to the patient's symptoms.

Route of Administration

Ophthalmic Use only.

Contraindications

Hypersensitivity to the active substance levofloxacin, to other quinolones or to any of the excipients.

Warnings and Precautions

Levofloxacin 5 mg/ml eye drops must not be injected sub-conjunctivally. The solution should not be introduced directly into the anterior chamber of the eye.

As with other anti-infectives, prolonged use may result in overgrowth of nonsusceptible organisms, including fungi. If worsening of infection occurs, or if a clinical improvement is not noted within a reasonable period, discontinue use and institute alternative therapy. Whenever clinical judgement dictates, the patient should be examined with the aid of magnification, such as slit-lamp biomicroscopy, and, where appropriate, fluorescein staining.

Systemic fluoroquinolones have been associated with hypersensitivity reactions, even following a single dose. If an allergic reaction to levofloxacin occurs, discontinue the medication.

Tendon inflammation and rupture may occur with systemic fluoroquinolone therapy including levofloxacin, particularly in older patients and those treated concurrently with corticosteroids. Therefore, caution should be exercised and treatment with Levofloxacin 5 mg/ml eye drops should be discontinued at the first sign of tendon inflammation.

Patients with external bacterial ocular infections should not wear contact lenses.

Paediatric population

The warnings and precautions for use are the same for adults and children aged ≥ 1 year.

Excipients

Levofloxacin 5 mg/ml eye drops contain benzalkonium chloride, which may be absorbed by soft contact lenses and may change the colour of the contact lenses. Contact lenses should be removed before using this medicine and may be put back in 15 minutes afterwards.

From the limited data available, there is no difference in the adverse event profile in children compared to adults. Generally, however, eyes in children show a stronger reaction for a given stimulus than the adult eye. Irritation may have an effect on treatment adherence in children.

Benzalkonium chloride has been reported to cause eye irritation, symptoms of dry eyes and may affect the tear film and corneal surface. Should be used with caution in dry eye patients and in patients where the cornea may be compromised.

Patients should be monitored in case of prolonged use.

Interactions with Other Medicaments

Specific drug interaction studies have not been conducted with levofloxacin eye drops.

Since maximum plasma concentrations of levofloxacin after ocular administration are at least 1000 times lower than those reported after standard oral doses, interactions mentioned for systemic use are unlikely to be clinically relevant when using Levofloxacin 5 mg/ml eye drops.

Pediatric population

No interaction studies have been performed.

Pregnancy and Lactation**Pregnancy**

There are no adequate data from the use of levofloxacin in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. The potential risk for humans is unknown. Levofloxacin 5 mg/ml eye drops should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.

Breastfeeding

Levofloxacin is excreted in human milk. However, at therapeutic doses of Levofloxacin no effects on the suckling child are anticipated. Levofloxacin 5 mg/ml eye drops should be used during lactation only if the potential benefit justifies any potential risk to the nursing child.

Fertility

Levofloxacin caused no impairment of fertility in rats at exposures considerably in excess of the maximum human exposure after ocular administration

Side Effects

Approximately 10 % of patients can be expected to experience adverse reactions. The reactions are usually graded as mild or moderate, are transient, and are generally restricted to the eye.

As the product contains benzalkonium chloride, contact eczema and/or irritation may be due to the active component or to this preservative.

The following undesirable effects assessed as definitely, probably or possibly related to treatment were reported during clinical trials and post-marketing experience with levofloxacin containing eye drops:

Frequency categories are expressed as: *Common* ($\geq 1/100$ to $< 1/10$); *Uncommon* ($\geq 1/1,000$ to $< 1/100$); *Rare* ($\geq 1/10,000$ to $< 1/1,000$); *Very rare* ($< 1/10,000$);

Immune system disorders

Rare: Extra-ocular allergic reactions, including skin rash

Very rare: Anaphylaxis

Nervous system disorders

Uncommon: Headache

Eye disorders

Common: Ocular burning, decreased vision and mucous strand.

Uncommon: Lid matting, chemosis, conjunctival papillary reaction, lid oedema, ocular discomfort, ocular itching, ocular pain, conjunctival injection, conjunctival follicles, ocular dryness, lid erythema, and photophobia.

No corneal precipitates were observed in clinical studies.

Respiratory, thoracic and mediastinal disorders

Uncommon: Rhinitis

Very rare: Laryngeal oedema

Additional adverse reactions that have been seen with the systemic use of the active substance (levofloxacin), and may potentially occur also with Levofloxacin 5mg/ml eye drops

Ruptures of the shoulder, hand, Achilles, or other tendons that required surgical repair or resulted in prolonged disability have been reported in patients receiving systemic fluoroquinolones. Studies and post marketing experience with systemic quinolones indicate that a risk of these ruptures may be increased in patients receiving corticosteroids, especially geriatric patients and in tendons under high stress, including Achilles tendon.

Pediatric population

Frequency, type and severity of adverse reactions in children are expected to be the same as in adults.

Symptoms and Treatment of Overdose

The total amount of levofloxacin in a bottle of eye drops is too small to induce toxic effects after an accidental oral intake. If considered necessary, the patient can be observed clinically and supportive measures can be undertaken. After a local overdose with Levofloxacin 5 mg/ml eye drops, the eyes can be flushed with clean (tap) water at room temperature.

Pediatric population

Actions to be taken in case of overdose are the same in adults and in children aged ≥ 1 year.

Effects on Ability to Drive and Use Machine

Levofloxan has minor influence on the ability to drive and use machines.

If there are any transient effects on vision, the patient should be advised to wait until this clear before driving or operating machinery.

Instructions for Use

User instruction: Be careful not to touch the tip of the bottle to the eye directly in order to avoid the contamination of the drug.

Method of administration

Before applying the drops:

If possible, ask someone else to apply the drops for you. Ask them to read these instructions with you before applying the drops.

- 1) Wash your hands.
- 2) Open the bottle. Take special care that the tip of the dropper bottle does not touch your eye, the skin around your eye or your fingers.
- 3) Tilt your head backwards and hold the bottle upside down over the eye.
- 4) Pull the lower eyelid downwards and look up. Gently squeeze the bottle and let one drop fall into the space between the lower eyelid and the eye.
- 5) Close the eye and press the inner corner of the eye with your finger for about one minute. Thus, you can prevent the eye drop from draining down the tear duct.
- 6) Wipe off any excess solution from the skin around the eye.
- 7) Put the cap back on and close the bottle tightly.

Levofloxan eye drops should not be injected into the interior part of the eyeball.

Storage Condition

Keep out of the sight and reach of children.

Store below 30° C. Keep bottle in the outer carton in order to protect from light.

Shelf Life 2 (two) years from date of manufacture. Shelf life after first opening: 28 days.

Dosage forms and packaging available 5 ml Balkanpharma Levofloxacin eye drops in white plastic bottles. One bottle per carton.

Name and address of manufacturer

Balkapharma-Razgrad AD

68 Aprilsko Vastanie Blvd Razgrad 7200, Bulgaria

Name and address of product registration holder

Image Pharma Sdn. Bhd.

10B, 2nd Floor, JalanSS3/5, Taman Sentosa, 47300 Petaling Jaya, Selangor, Malaysia

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Foot note: font size: 7, as measure in Times New Roman